

ECE 0102 Chapter 1 Problems 1.35–1.62 Solutions

放大器、频率响应、元件容差与有效数字

核心规则. 对正弦信号

$$v(t) = V_m \sin(\omega t + \phi),$$

幅值是 V_m , 相位是 ϕ , 频率是 $f = \omega/(2\pi)$ 。电压增益用相量比值:

$$A_v = \frac{V_o \angle \phi_o}{V_s \angle \phi_s}, \quad |A_v| = \frac{V_o}{V_s}, \quad \angle A_v = \phi_o - \phi_s.$$

多频率信号不能整体相除, 必须按频率分量逐项处理。

Problem 1.35

Given

$$v_s = 10^{-4} \sin(2 \times 10^7 \pi t) \text{ V}, \quad v_o = 4 \sin(2 \times 10^7 \pi t + 56^\circ) \text{ V}.$$

Both signals have the same angular frequency, so the gain can be found by comparing amplitudes and phases.

$$|A_v| = \frac{4}{10^{-4}} = 4.0 \times 10^4.$$

The input phase is 0° and the output phase is 56° , hence

$$\angle A_v = 56^\circ - 0^\circ = 56^\circ.$$

$A_v = 4.0 \times 10^4 \angle 56^\circ$

Problem 1.36

Given

$$v_s = 10^{-3} \sin(3000\pi t) + 2 \times 10^{-3} \sin(5000\pi t) \text{ V},$$

$$v_o = 10^{-2} \sin(3000\pi t - 45^\circ) + 10^{-1} \sin(5000\pi t - 12^\circ) \text{ V}.$$

First convert each angular frequency into ordinary frequency:

$$f = \frac{\omega}{2\pi}.$$

For $3000\pi t$, $\omega = 3000\pi$, so

$$f = \frac{3000\pi}{2\pi} = 1500 \text{ Hz}.$$

For $5000\pi t$, $\omega = 5000\pi$, so

$$f = \frac{5000\pi}{2\pi} = 2500 \text{ Hz}.$$

(a) At 2500 Hz:

$$A_v(2500 \text{ Hz}) = \frac{10^{-1} \angle (-12^\circ)}{2 \times 10^{-3} \angle 0^\circ} = 50 \angle (-12^\circ).$$

$|A_v| = 50, \angle A_v = -12^\circ$

(b) At 1500 Hz:

$$A_v(1500 \text{ Hz}) = \frac{10^{-2} \angle(-45^\circ)}{10^{-3} \angle 0^\circ} = 10 \angle(-45^\circ).$$

$$\boxed{|A_v| = 10, \angle A_v = -45^\circ}$$

Problem 1.37

Figure 1.20 is an inverting amplifier, so

$$A_v = \frac{v_o}{v_s} = -\frac{R_2}{R_1}.$$

The minus sign means a 180° phase inversion.

(a) $R_1 = 14 \text{ k}\Omega$, $R_2 = 560 \text{ k}\Omega$:

$$A_v = -\frac{560}{14} = -40.$$

(b) $R_1 = 18 \text{ k}\Omega$, $R_2 = 360 \text{ k}\Omega$:

$$A_v = -\frac{360}{18} = -20.$$

(c) $R_1 = 1.8 \text{ k}\Omega$, $R_2 = 62 \text{ k}\Omega$:

$$A_v = -\frac{62}{1.8} = -34.4.$$

$$\boxed{(a) -40, \quad (b) -20, \quad (c) -34.4}$$

Problem 1.38

For the same inverting amplifier,

$$v_o(t) = -\frac{R_2}{R_1} v_s(t).$$

Given

$$R_1 = 910 \Omega, \quad R_2 = 7.5 \text{ k}\Omega, \quad v_s(t) = 0.01 \sin(750\pi t) \text{ V}.$$

Thus

$$\frac{R_2}{R_1} = \frac{7500}{910} = 8.2418.$$

So

$$v_o(t) = -(8.2418)(0.01 \sin(750\pi t)) = -0.0824 \sin(750\pi t) \text{ V}.$$

The source current through R_1 is

$$i_s(t) = \frac{v_s(t)}{R_1} = \frac{0.01}{910} \sin(750\pi t) = 1.099 \times 10^{-5} \sin(750\pi t) \text{ A}.$$

$$\boxed{v_o(t) = -0.0824 \sin(750\pi t) \text{ V}}$$

$$\boxed{i_s(t) = 11.0 \sin(750\pi t) \mu\text{A}}$$

Problem 1.39

The circuit is a voltage follower. For an ideal op amp with negative feedback,

$$v_- = v_+.$$

Here $v_+ = v_i$ and $v_- = v_o$, therefore

$$v_o = v_i.$$
$$\boxed{A_v = \frac{v_o}{v_i} = 1}$$

Problem 1.40

This is a non-inverting amplifier. For an ideal op amp,

$$v_- = v_+ = v_i.$$

The feedback divider gives

$$v_i = v_o \frac{R_1}{R_1 + R_2}.$$

Therefore

$$\frac{v_o}{v_i} = \frac{R_1 + R_2}{R_1} = 1 + \frac{R_2}{R_1}.$$

$$\boxed{A_v = 1 + \frac{R_2}{R_1}}$$

Problem 1.41

The non-inverting input is grounded, so the inverting node is a virtual ground:

$$v_- = v_+ = 0.$$

Apply KCL at the inverting node:

$$\frac{v_1 - 0}{R_1} + \frac{v_2 - 0}{R_2} = \frac{0 - v_o}{R_3}.$$

Hence

$$v_o = -\frac{R_3}{R_1}v_1 - \frac{R_3}{R_2}v_2.$$

Given

$$R_1 = 2 \text{ k}\Omega, \quad R_2 = 10 \text{ k}\Omega, \quad R_3 = 51 \text{ k}\Omega,$$

$$v_1(t) = 0.01 \sin(3770t) \text{ V}, \quad v_2(t) = 0.05 \sin(10000t) \text{ V}.$$

Thus

$$\frac{R_3}{R_1} = \frac{51}{2} = 25.5, \quad \frac{R_3}{R_2} = \frac{51}{10} = 5.1.$$

$$v_o(t) = -(25.5)(0.01 \sin(3770t)) - (5.1)(0.05 \sin(10000t)).$$

$$\boxed{v_o(t) = -0.255 \sin(3770t) - 0.255 \sin(10000t) \text{ V}}$$

The voltage at the inverting node is the virtual ground:

$$\boxed{v_-(t) = 0 \text{ V.}}$$

Problem 1.42

This circuit is an inverting summing DAC. Each switch connects the corresponding input resistor either to 0 V or to V_{REF} . For an ideal op amp, the summing node is at virtual ground. Therefore

$$v_o = -R \left(\frac{b_1 V_{\text{REF}}}{2R} + \frac{b_2 V_{\text{REF}}}{4R} + \frac{b_3 V_{\text{REF}}}{8R} \right),$$

so

$$v_o = -V_{\text{REF}} \left(\frac{b_1}{2} + \frac{b_2}{4} + \frac{b_3}{8} \right).$$

(a) For $(b_1 b_2 b_3) = (011)$ and $V_{\text{REF}} = 5.0 \text{ V}$:

$$v_o = -5 \left(0 + \frac{1}{4} + \frac{1}{8} \right) = -5 \left(\frac{3}{8} \right) = -1.875 \text{ V}.$$

(b) For $(b_1 b_2 b_3) = (100)$:

$$v_o = -5 \left(\frac{1}{2} + 0 + 0 \right) = -2.500 \text{ V}.$$

(c) Table of all possible input combinations:

Input $b_1 b_2 b_3$	v_o (V)
000	0
001	-0.625
010	-1.250
011	-1.875
100	-2.500
101	-3.125
110	-3.750
111	-4.375

Problem 1.43

The amplifier has zero gain below 1000 Hz, gain 20 from 1000 Hz to 5000 Hz, and zero gain above 5000 Hz. Only a finite middle frequency band passes. Therefore it is a band-pass amplifier.

Band-pass amplifier

Problem 1.44

The amplifier has gain 10 below 6000 Hz and zero gain above 6000 Hz. Low frequencies pass; high frequencies are rejected.

Low-pass amplifier

Problem 1.45

The amplifier in Problem 1.44 has gain 10 for frequencies below 6000 Hz, and zero gain above 6000 Hz.

$$v_s(t) = 5 \sin(2000\pi t) + 3 \cos(8000\pi t) + 2 \cos(15000\pi t).$$

Find each frequency:

$$f_1 = \frac{2000\pi}{2\pi} = 1000 \text{ Hz}, \quad f_2 = \frac{8000\pi}{2\pi} = 4000 \text{ Hz}, \quad f_3 = \frac{15000\pi}{2\pi} = 7500 \text{ Hz}.$$

The 1000 Hz and 4000 Hz terms pass with gain 10. The 7500 Hz term is rejected.

$$v_o(t) = 50 \sin(2000\pi t) + 30 \cos(8000\pi t) \text{ V}$$

Problem 1.46

The amplifier has zero gain for frequencies below 10 kHz and gain 16 for frequencies above 10 kHz. High frequencies pass; low frequencies are rejected.

$$\text{High-pass amplifier}$$

Problem 1.47

The amplifier in Problem 1.43 is a band-pass amplifier with gain 20 between 1000 Hz and 5000 Hz, zero otherwise.

$$v_s(t) = 0.5 \sin(2500\pi t) + 0.75 \cos(8000\pi t) + 0.6 \cos(12000\pi t).$$

Frequencies:

$$f_1 = \frac{2500\pi}{2\pi} = 1250 \text{ Hz}, \quad f_2 = \frac{8000\pi}{2\pi} = 4000 \text{ Hz}, \quad f_3 = \frac{12000\pi}{2\pi} = 6000 \text{ Hz}.$$

The first two terms lie in the passband; the third is above 5000 Hz and is rejected.

$$v_o(t) = 10.0 \sin(2500\pi t) + 15.0 \cos(8000\pi t) \text{ V}$$

Problem 1.48

The amplifier in Problem 1.46 is high-pass with gain 16 only above 10 kHz.

$$v_s(t) = 0.5 \sin(2500\pi t) + 0.75 \cos(8000\pi t) + 0.8 \cos(12000\pi t).$$

Frequencies:

$$1250 \text{ Hz}, \quad 4000 \text{ Hz}, \quad 6000 \text{ Hz}.$$

All are below 10 kHz, so all are rejected.

$$v_o(t) = 0$$

Problem 1.49

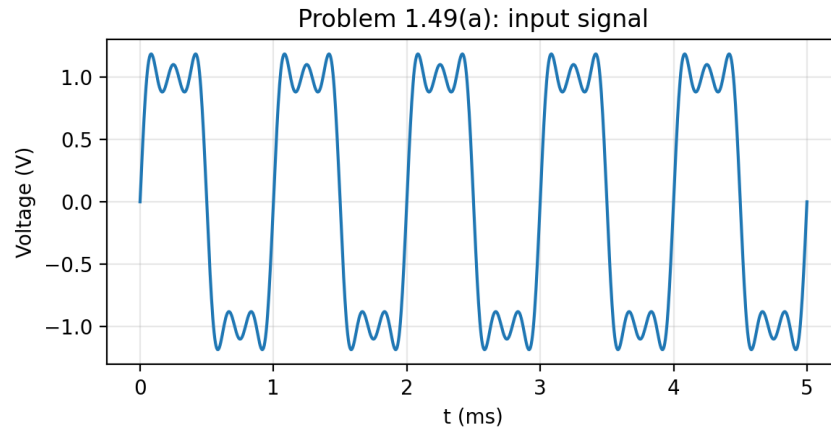
Given

$$v(t) = \frac{4}{\pi} \left(\sin \omega_0 t + \frac{1}{3} \sin 3\omega_0 t + \frac{1}{5} \sin 5\omega_0 t \right) \text{ V}, \quad f_0 = 1000 \text{ Hz}.$$

Thus

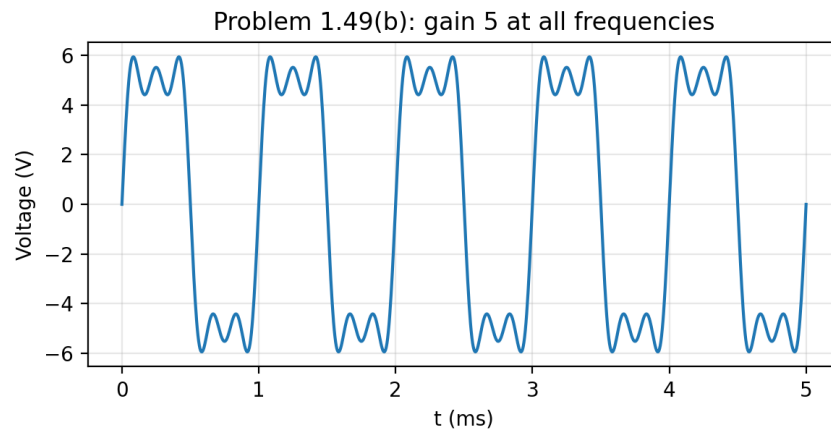
$$\omega_0 = 2\pi f_0 = 2000\pi \text{ rad/s}.$$

(a) **Input signal plot.**



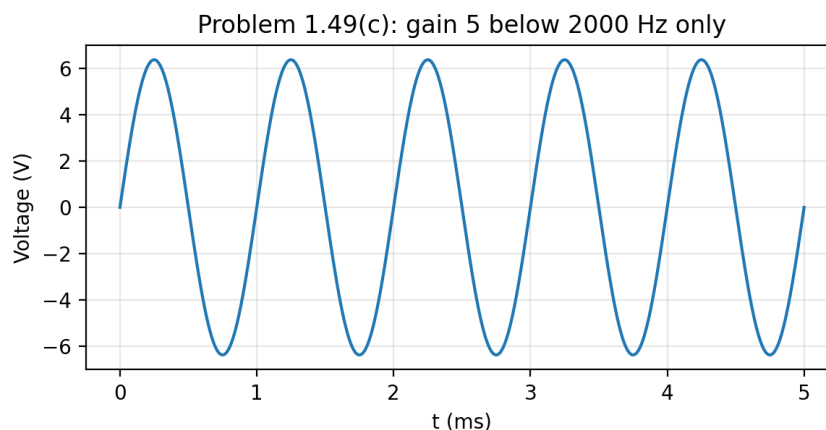
(b) **Gain of 5 at all frequencies.** Every frequency component is multiplied by 5:

$$v_o(t) = \frac{20}{\pi} \left(\sin \omega_0 t + \frac{1}{3} \sin 3\omega_0 t + \frac{1}{5} \sin 5\omega_0 t \right) \text{ V}.$$



(c) **Gain of 5 below 2000 Hz and zero above 2000 Hz.** The components are at 1000 Hz, 3000 Hz, and 5000 Hz. Only the 1000 Hz component remains:

$$v_o(t) = \frac{20}{\pi} \sin \omega_0 t \text{ V}.$$

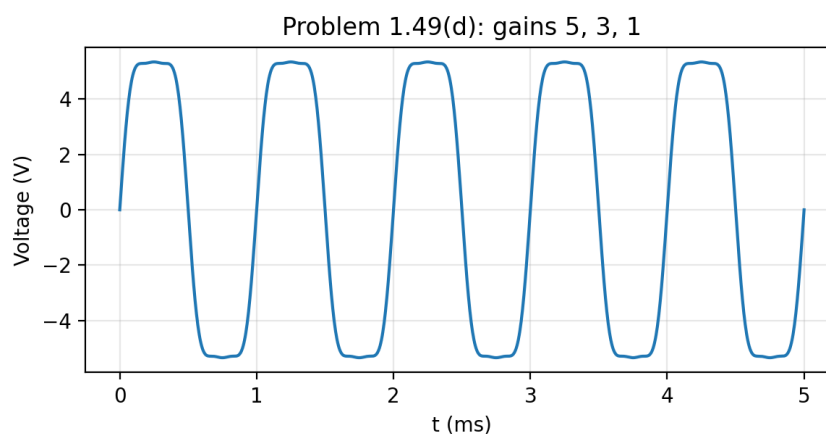


(d) **Gains: 5 at 1000 Hz, 3 at 3000 Hz, and 1 at 5000 Hz.**

$$v_o(t) = \frac{4}{\pi} \left[5 \sin \omega_0 t + 3 \left(\frac{1}{3} \sin 3\omega_0 t \right) + 1 \left(\frac{1}{5} \sin 5\omega_0 t \right) \right] \text{ V.}$$

Therefore

$$v_o(t) = \frac{4}{\pi} \left(5 \sin \omega_0 t + \sin 3\omega_0 t + \frac{1}{5} \sin 5\omega_0 t \right) \text{ V}$$



Problem 1.50

For a nominal resistor $R_N = 4.7 \text{ k}\Omega = 4700\Omega$, tolerance means

$$R_{\min} = R_N(1 - T), \quad R_{\max} = R_N(1 + T).$$

(a) **1 percent:**

$$R_{\min} = 4700(0.99) = 4653\Omega, \quad R_{\max} = 4700(1.01) = 4747\Omega.$$

(b) **5 percent:**

$$R_{\min} = 4700(0.95) = 4465\Omega, \quad R_{\max} = 4700(1.05) = 4935\Omega.$$

(c) 10 percent:

$$R_{\min} = 4700(0.90) = 4230\Omega, \quad R_{\max} = 4700(1.10) = 5170\Omega.$$

$1\% : 4653 - -4747\Omega; \quad 5\% : 4465 - -4935\Omega; \quad 10\% : 4230 - -5170\Omega$

Problem 1.51

Nominal capacitance:

$$C_N = 10000 \mu\text{F}.$$

Tolerance is asymmetric: $+20\% / -50\%$.

$$C_{\min} = 10000(1 - 0.50) = 5000 \mu\text{F},$$

$$C_{\max} = 10000(1 + 0.20) = 12000 \mu\text{F}.$$

$5000 \mu\text{F} \leq C \leq 12000 \mu\text{F}$
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Problem 1.52

Nominal voltage:

$$V_N = 1.8 \text{ V}.$$

Maximum allowed deviation:

$$\Delta V = 50 \text{ mV} = 0.050 \text{ V}.$$

Tolerance percentage:

$$T = \frac{\Delta V}{V_N} \times 100\% = \frac{0.050}{1.8} \times 100\% = 2.78\%.$$

Tolerance = $\pm 2.78\%$

Problem 1.53

Nominal resistor:

$$R_N = 8200\Omega, \quad T = 10\%.$$

Allowed range:

$$R_{\min} = 8200(0.90) = 7380\Omega, \quad R_{\max} = 8200(1.10) = 9020\Omega.$$

Measured value is 7905Ω , and

$$7380\Omega < 7905\Omega < 9020\Omega.$$

The resistor is within specification.

Problem 1.54

(a) A 5 V supply with 5% tolerance has allowed range

$$V_{\min} = 5(0.95) = 4.75 \text{ V}, \quad V_{\max} = 5(1.05) = 5.25 \text{ V}.$$

Measured output is 5.30 V, which is above 5.25 V.

Based only on the reading, it is outside specification.

(b) The voltmeter has 1.5% tolerance. If the meter reads high by 1.5%, then

$$V_{\text{actual}} = \frac{5.30}{1.015} = 5.22 \text{ V},$$

which is inside the specification range. If the meter reads low by 1.5%, then

$$V_{\text{actual}} = \frac{5.30}{0.985} = 5.38 \text{ V},$$

which is outside the range.

So the meter tolerance changes the certainty of the conclusion: the supply may be acceptable if the meter is reading high, but the measurement is not accurate enough to prove that it is acceptable.

With meter tolerance included, the result is inconclusive without a better measurement.

Problem 1.55

Given

$$R(0^\circ\text{C}) = 6066\Omega, \quad R(100^\circ\text{C}) = 6562\Omega.$$

Temperature coefficient in $\Omega/^\circ\text{C}$:

$$\text{TCR} = \frac{\Delta R}{\Delta T} = \frac{6562 - 6066}{100 - 0} = 4.96\Omega/^\circ\text{C}.$$

At $T_{\text{NOM}} = 27^\circ\text{C}$:

$$R_{\text{nom}} = 6066 + 4.96(27) = 6199.92\Omega \approx 6200\Omega.$$

If expressed as a relative coefficient around the nominal value:

$$\alpha = \frac{4.96}{6200} = 8.00 \times 10^{-4}/^\circ\text{C} = 800 \text{ ppm}/^\circ\text{C}.$$

$$\text{TCR} = 4.96\Omega/^\circ\text{C}, \quad R_{\text{nom}} \approx 6.20 \text{ k}\Omega$$

Problem 1.56

This problem uses the current-divider circuit from Problem 1.22. Let

$$I = 5 \text{ mA}, \quad R_1 = 2.4 \text{ k}\Omega, \quad R_2 = 5.6 \text{ k}\Omega, \quad R_3 = 3.6 \text{ k}\Omega.$$

The two branch currents are

$$I_1 = I \frac{R_2 + R_3}{R_1 + R_2 + R_3}, \quad I_2 = I \frac{R_1}{R_1 + R_2 + R_3},$$

and

$$V_3 = I_2 R_3.$$

Current source tolerance is $\pm 2\%$ and each resistor tolerance is $\pm 5\%$.

Worst-case values are obtained by pushing each variable in the direction that maximizes or minimizes the target expression.

$$I_{\max} = 5(1.02) = 5.10 \text{ mA}, \quad I_{\min} = 5(0.98) = 4.90 \text{ mA}.$$

$$R_{1\max} = 2.4(1.05) = 2.52 \text{ k}\Omega, \quad R_{1\min} = 2.4(0.95) = 2.28 \text{ k}\Omega,$$

$$R_{2\max} = 5.6(1.05) = 5.88 \text{ k}\Omega, \quad R_{2\min} = 5.6(0.95) = 5.32 \text{ k}\Omega,$$

$$R_{3\max} = 3.6(1.05) = 3.78 \text{ k}\Omega, \quad R_{3\min} = 3.6(0.95) = 3.42 \text{ k}\Omega.$$

For I_1 , make I , R_2 , R_3 large and R_1 small for maximum:

$$I_{1\max} = 5.10 \frac{5.88 + 3.78}{2.28 + 5.88 + 3.78} = 4.12 \text{ mA}.$$

For minimum:

$$I_{1\min} = 4.90 \frac{5.32 + 3.42}{2.52 + 5.32 + 3.42} = 3.80 \text{ mA}.$$

For I_2 , make I and R_1 large while R_2, R_3 are small for maximum:

$$I_{2\max} = 5.10 \frac{2.52}{2.52 + 5.32 + 3.42} = 1.14 \text{ mA}.$$

For minimum:

$$I_{2\min} = 4.90 \frac{2.28}{2.28 + 5.88 + 3.78} = 0.936 \text{ mA}.$$

For $V_3 = I_2 R_3$, direct substitution gives the worst-case bounds:

$$V_{3\max} = 4.18 \text{ V}, \quad V_{3\min} = 3.30 \text{ V}.$$

$I_1 = 3.80 \text{ to } 4.12 \text{ mA}, \quad I_2 = 0.936 \text{ to } 1.14 \text{ mA}, \quad V_3 = 3.30 \text{ to } 4.18 \text{ V}$
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Problem 1.57

This problem uses the circuit from Problem 1.20. The nominal values are

$$V = 10 \text{ V}, \quad R_1 = 22 \text{ k}\Omega, \quad R_2 = 47 \text{ k}\Omega, \quad R_3 = 180 \text{ k}\Omega.$$

The parallel combination is

$$R_X = R_2 \parallel R_3 = \frac{R_2 R_3}{R_2 + R_3}.$$

Then

$$V_1 = V \frac{R_1}{R_1 + R_X}.$$

The source current is

$$I = \frac{V}{R_1 + R_X}.$$

Current division in the parallel branch gives

$$I_2 = I \frac{R_3}{R_2 + R_3}, \quad I_3 = I \frac{R_2}{R_2 + R_3}.$$

Voltage source tolerance is $\pm 5\%$ and resistor tolerances are $\pm 10\%$.

Using worst-case substitution:

$$\boxed{V_{1\min} = 3.09 \text{ V}, \quad V_{1\max} = 4.40 \text{ V}}$$

$$\boxed{I_{2\min} = 114 \mu\text{A}, \quad I_{2\max} = 158 \mu\text{A}}$$

$$\boxed{I_{3\min} = 28.3 \mu\text{A}, \quad I_{3\max} = 43.1 \mu\text{A}}$$

The key point: worst-case analysis does not mean all components move in the same direction. Each target quantity has its own maximizing/minimizing combination.

Problem 1.58

For the Thevenin equivalent resistance from Problem 1.25,

$$R_{\text{th}} = \left(g_m + \frac{1}{R_1} \right)^{-1}.$$

Nominal values are

$$g_m = 0.002 \text{ S}, \quad R_1 = 50 \text{ k}\Omega.$$

Both have $\pm 20\%$ tolerance.

To maximize R_{th} , minimize the denominator: use g_m minimum and R_1 maximum.

$$R_{\text{th,max}} = \left[0.002(0.8) + \frac{1}{50,000(1.2)} \right]^{-1} = 619\Omega.$$

To minimize R_{th} , maximize the denominator: use g_m maximum and R_1 minimum.

$$R_{\text{th,min}} = \left[0.002(1.2) + \frac{1}{50,000(0.8)} \right]^{-1} = 412\Omega.$$

$$\boxed{412\Omega \leq R_{\text{th}} \leq 619\Omega}$$

Problem 1.59

A 200-case Monte Carlo analysis was performed for the circuit in Problem 1.56 using uniform random variations within the specified tolerances. The values below are from one reproducible run with seed 102.

Quantity	Monte Carlo min	Monte Carlo max	Monte Carlo mean	Worst-case min	Worst-case max
I_1	3.861 mA	4.086 mA	3.962 mA	3.80 mA	4.12 mA
I_2	0.952 mA	1.120 mA	1.034 mA	0.936 mA	1.14 mA
V_3	3.426 V	4.040 V	3.737 V	3.30 V	4.18 V

The Monte Carlo range is narrower than the worst-case range. This is expected: worst-case assumes all variables simultaneously take the most damaging values, while Monte Carlo randomly samples combinations, so extreme simultaneous alignment is unlikely in only 200 cases.

Problem 1.60

A 200-case Monte Carlo analysis was performed for the circuit in Problem 1.57 using uniform random variations within the specified tolerances. The values below are from one reproducible run with seed 102.

Quantity	Monte Carlo min	Monte Carlo max	Monte Carlo mean	Worst-case min	Worst-case max
V_1	3.190 V	4.278 V	3.711 V	3.09 V	4.40 V
I_2	118.4 μ A	152.2 μ A	133.1 μ A	114 μ A	158 μ A
I_3	29.61 μ A	39.74 μ A	34.67 μ A	28.3 μ A	43.1 μ A

Again, the Monte Carlo results stay inside the worst-case limits. Worst-case analysis is deliberately conservative; Monte Carlo gives a more realistic sample of expected spread but cannot guarantee absolute limits unless the sample space is fully exhausted.

Problem 1.61

(a) **Three significant digits:**

$$3.2947 \rightarrow 3.29, \quad 0.995171 \rightarrow 0.995, \quad -6.1551 \rightarrow -6.16.$$

$$\boxed{3.29, \quad 0.995, \quad -6.16}$$

(b) **Four significant digits:**

$$3.2947 \rightarrow 3.295, \quad 0.995171 \rightarrow 0.9952, \quad -6.1551 \rightarrow -6.155.$$

$$\boxed{3.295, \quad 0.9952, \quad -6.155}$$

(c) A calculator check gives the same rounding if standard significant-digit rules are applied. The sign does not count as a significant digit; leading zeros also do not count.

Problem 1.62

Use Ohm's law:

$$V = IR.$$

(a)

$$I = 1.763 \text{ mA}, \quad R = 20.70 \text{ k}\Omega.$$

Since $\text{mA} \cdot \text{k}\Omega = \text{V}$,

$$V = (1.763)(20.70) = 36.4941 \text{ V}.$$

Three significant digits:

$$\boxed{V = 36.5 \text{ V}}$$

(b) Two significant digits:

$$36.4941 \text{ V} \rightarrow \boxed{36 \text{ V}}.$$

(c)

$$I = 102.1 \mu\text{A}, \quad R = 97.80 \text{ k}\Omega.$$

$$V = (102.1 \times 10^{-6})(97.80 \times 10^3) = 9.986 \text{ V}.$$

Three significant digits:

$$\boxed{9.99 \text{ V}}$$

Two significant digits:

$$\boxed{10 \text{ V}}$$